

ENGINEERING WORKSHOP PRACTICES

I Semester (AERO, MECH & EEE)								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6ME03	ESC	L	T	P	C	CIE	SEE	Total
		0	1	2	2	40	60	100
COURSEOBJECTIVES:								
1. To Study of different hand operated tools, uses and their demonstration. 2. To gain a good basic working knowledge required for the production of various engineering products. 3. To provide hands-on experience about use of different engineering materials, tools, equipments and processes those are common in the engineering field. 4. It explains the construction, function, use and application of different working tools, equipment and machines. 5. To study commonly used carpentry joints. 6. Identify and use marking out tools, hand tools, measuring equipment and to work to prescribed tolerances. 7. To develop a right attitude, team working, precision and safety at workplace.								
WEEKS	TRADES FOR EXERCISES							
	Fitting							
Week 1	Filing Four Sides of MS Work piece							
Week 2	L Fit							
	Carpentry							
Week 3	Half Lap Joint							
Week 4	Dove Tail Joint							
	Tin Smithy							
Week 5	Prepare a Rectangular Tray							
Week 6	Prepare A Square Tin							
	Electrical							
Week 7	Series and Parallel Connection One Way Switch							
Week 8	Two Way Switch Connection Stair Case Wiring							
	Electronics							
Week 9	Soldering - Series Connection & Parallel Connection							
Week 10	Desoldering & Construction of Wheat stone bridge							
	Black Smithy							
Week 11	Blacksmithy : Round to S-Hook							
Week 12	Blacksmithy : Round to Chisel							
	TRADES FOR DEMONSTRATION & EXPOSURE							
Week 13	Introduction to Plumbing							

TEXT BOOKS

1. Work shop Manual - P. Kannaiyah/ K.L. Narayana/ Scitech.
2. Workshop Manual / K. Venugopal / Anuradha Publications.

REFERENCE BOOKS:

1. Hajra Choudhury S.K., Hajra Choudhury A.K. and Nirjhar Roy S.K., "Elements of Workshop Technology", Media promoters and publishers private limited, Mumbai, Vol. I 2008 and Vol. II 2010.
2. Kalpakjian S. and Steven S. Schmid, "Manufacturing Engineering and Technology", Pearson Education India Edition, 4th edition, 2002.

COURSE OUTCOMES:**STUDENTS SHOULD BE ABLE TO:**

1. Assemble different components.
2. Identify and apply suitable tools for different trades of Engineering processes.
3. Practice on manufacturing of components using workshop trades including Soldering, Carpentry, House wiring, Carpentry, Fitting and Tin smithy & Fabricate Components with their own hands.
4. Apply basic electrical engineering knowledge for house wiring practice.
5. Learn the safety precautions for various operations in basic trade operations.